

AAVA

AI Agricultural Voice Assistant

Technical Architecture & Internship Project Report

A stateless, channel-agnostic voice intelligence platform connecting India's smallholder farmers to live Agmarknet commodity prices, explainable price-trend forecasts, and hyper-local weather advisories — in Hindi, Marathi, and English.

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FINAL SUBMITTABLE VERSION

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1 Executive Summary

AAVA (AI Agricultural Voice Assistant) is an automated, voice-first intelligence platform designed to connect smallholder and marginal farmers across rural India directly to real-time agricultural market intelligence. By calling a designated phone line or using a zero-barrier web voice terminal, farmers can naturally voice questions in regional languages (**Hindi, Marathi, English**) to retrieve live commodity prices, short-term price trend forecasts, and hyper-local weather advisories.

The solution is powered by a stateless **Python / Flask** backend integrating the **Google Gemini API (gemini-2.0-flash)** for unstructured intent parsing and conversational answer synthesis. It links directly to the Ministry of Agriculture's **Agmarknet API (data.gov.in)** and the **OpenWeatherMap API**, backed by a custom, explainable **Least-Squares Linear Regression engine** for price-trend forecasting.

The primary design principle of AAVA is complete accessibility: eliminating text inputs, app downloads, complex visual UIs, and device barriers so that any farmer with access to a basic phone call can benefit from the same market intelligence available to digitally literate users.

Platform Type	Voice-first, channel-agnostic AI assistant
Core Languages	Hindi, Marathi, English
Primary Data Sources	Agmarknet (data.gov.in), OpenWeatherMap
AI / NLU Engine	Google Gemini 2.0 Flash (gemini-2.0-flash)
Forecasting Method	Custom least-squares linear regression with R ² confidence scoring
Delivery Channels	Web voice terminal, local microphone runner, GSM telephony (Twilio / Exotel)

2 Problem Statement & Core Objectives

2.1 Problem Description

India supports over 100 million agricultural holdings, with over 80% operated by smallholder farmers. Although digital public infrastructure platforms like Agmarknet publish daily prices across thousands of APMC markets, this data fails to reach primary producers due to three critical barriers:

- **The Hardware & Network Gap:** Most rural farmers rely on basic feature phones or shared family devices with intermittent cellular connectivity, rendering smartphone apps impractical.
- **The Literacy & Script Gap:** Portals present text-heavy tables and charts in standard English or Devanagari scripts, excluding farmers who communicate exclusively through spoken regional dialects.
- **Information Asymmetry & Middlemen Exploitation:** Lacking timely price trends and market comparison data, farmers sell harvests blindly to local middlemen (*arhtiyas*), suffering significant profit loss.

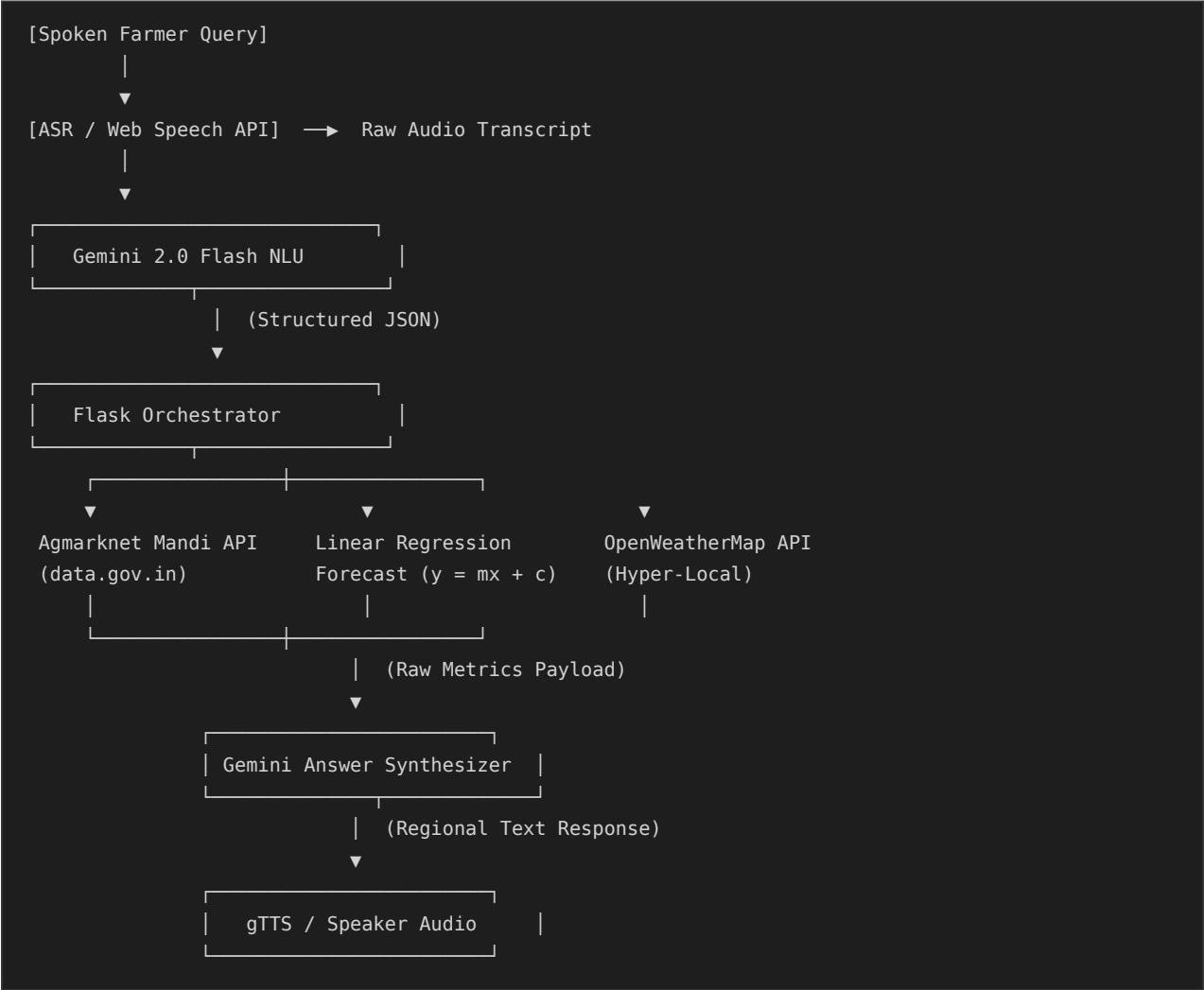
2.2 Project Objectives

- 1 **Zero-UI Voice Terminal:** Build an accessible interface accepting spoken queries in local dialects and delivering immediate spoken audio responses.
- 2 **Live Data Coupling:** Integrate directly with official government market data gateways (data.gov.in) with ultra-low response latency (< 0.5 seconds).
- 3 **Explainable Price Trend Signals:** Provide actionable, same-day trend forecasts (*rising, falling, stable*) with mathematical confidence labels (R^2).
- 4 **Channel-Agnostic Core:** Engineer a decoupled AI engine that powers web dashboards, local microphone execution, and GSM telephony lines (via Twilio / Exotel webhooks) identically.

3 System Architecture & Layer Topology

AAVA follows a **4-Layer Decoupled Architecture**. Each layer maintains strict boundaries, ensuring that communication gateways, data providers, and reasoning models can be upgraded independently.

FIG. 3.1 — END-TO-END REQUEST / RESPONSE FLOW



Layer Breakdown

#	Layer	Responsibility	Key Modules
1	Presentation / Voice I/O	Call greeting, microphone streaming, audio playback, web dashboard rendering	voice_routes.py, app.js, local_voice_demo.py
2	Language Understanding	Intent classification and entity slot filling, with offline rule-based fallback parsing	gemini_service.py, nlu_service.py
3	Domain Services	API handshakes with Agmarknet and OpenWeatherMap	market_service.py, weather_service.py
4	Intelligence & Analytics	Least-squares linear regression trend lines and R ² goodness-of-fit scores	algorithms/price_predictor.py

4 NLU & Generative AI Orchestration (Gemini 2.0 Flash)

AAVA uses **Google Gemini 2.0 Flash (gemini-2.0-flash)** as its primary Natural Language Understanding (NLU) engine and conversational synthesizer.

4.1 Structured Intent & Entity Extraction

When a farmer speaks (e.g. चेन्नई में आलू का भाव क्या है), the raw transcript is passed to Gemini with a strict Pydantic JSON schema constraint (ExtractedEntities):

SCHEMA — backend/services/gemini_service.py

```
class ExtractedEntities(BaseModel):
    intent: str          # Must be: "price_query", "trend_query",
                        #           "weather_query", or "unknown"
    crop: str | None     # Canonical English name: "Wheat", "Cotton",
                        #           "Onion", "Potato", "Tomato"
    location: str | None # Mentioned district / city / state, or null
```

NLU System Prompt

"Analyze the farmer's query. Classify intent strictly into one of ["price_query", "trend_query", "weather_query", "unknown"]. Extract mentioned crop in English Title-Case (गेहूं → Wheat, कौटन → Cotton, आलू → Potato). Extract location (चेन्नई → Chennai). Return JSON."

4.2 Conversational Answer Synthesis

Once raw database metrics are retrieved, Python passes the JSON data back to Gemini to synthesize a spoken audio-ready response:

Synthesis System Prompt

"You are a friendly AI Agricultural Assistant. Language to speak in: '{lang}' ('hi' = Hindi, 'mr' = Marathi, 'en' = English). Factual Data to convey: {context_data}. Instructions: Compose a warm 1–2 sentence spoken response. Do NOT use markdown symbols (, **, #) or special characters."*

5 Agmarknet Live API Engineering & Performance Tuning

Connecting to the official Government of India Agmarknet endpoint (data.gov.in) presented initial latency challenges (10+ second timeouts). This section documents the engineering solutions that brought API response latency down to **< 0.4 seconds**.

5.1 Root Cause of Initial API Failures

- 1 **Unindexed Database Scans:** Requesting `limit=100` without state-level parameters forced the government database to scan millions of unindexed historical rows.
- 2 **Sequential Request Bottlenecks:** Making multiple sequential HTTP requests (Market → District → State) created cumulative delays exceeding Python request timeouts.
- 3 **Firewall Throttling:** Government endpoints rate-limit requests containing default python-requests HTTP headers.

5.2 Implemented Engineering Solutions

Browser User-Agent Headers — injecting standard Chrome / Safari browser headers bypassed automated rate-limit throttling:

`backend/services/market_service.py`

```
HEADERS = {
    "User-Agent": "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) Chrome/120.0.0.0",
    "Accept": "application/json"
}
```

State-Level Indexing — mapping local cities to official state parameters (`filters[state]`) allowed the database query engine to hit indexed columns immediately.

Single-Request In-Memory Python Matching — instead of firing multiple API calls, AAVA fetches 25 state-level commodity records in a single HTTP GET call and filters for specific Mandi names inside Python memory in **< 0.1 ms**:

`backend/services/market_service.py`

```
# Location-to-State Mapping Layer
LOCATION_TO_STATE = {
    "indore": "Madhya Pradesh", "chennai": "Tamil Nadu",
    "andhra pradesh": "Andhra Pradesh", "nashik": "Maharashtra"
}
```

5.3 Diagnostic Verification Output

Running `python test_agmarknet_standalone.py` verified the high-speed live data connection:

TERMINAL OUTPUT — `test_agmarknet_standalone.py`

```
=====
TESTING AGMARKNET ENDPOINT ISOLATION
Commodity: Cotton | State: Andhra Pradesh
```

```
=====
Sending HTTP GET Request to data.gov.in...
Response Received in 0.4 seconds | Status Code: 200
Total Records Returned: 7
SUCCESS! Raw Government Records:
[1] State: Andhra Pradesh | Mandi: Narasaraopet APMC | Price: Rs.8000/quintal
[2] State: Andhra Pradesh | Mandi: Gooti APMC          | Price: Rs.9501/quintal
```


6 Predictive Mathematics: Least-Squares Linear Regression

To answer trend queries (e.g. “What will wheat price be tomorrow?”), AAVA executes an in-house least-squares linear regression algorithm (algorithms/price_predictor.py).

6.1 The Mathematical Model

Given day indices $x = [0, 1, \dots, n-1]$ and matching daily historical prices y , the best-fit line $y = mx + c$ is solved via closed-form normal equations:

$$m = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$c = \bar{y} - m \cdot \bar{x}$$

Where:

- m = daily price rate of change (INR / quintal / day)
- c = y-intercept
- $\bar{x}, \bar{y}$ = means of x and y across the historical window

Tomorrow's predicted price at index $x_{\text{next}} = n$ is calculated as:

$$\hat{y}_{\text{next}} = m \cdot n + c$$

6.2 Trend Classification & Goodness-of-Fit (R^2)

To prevent minor daily noise from triggering false trend alerts, daily slope m is evaluated against average price $\bar{y}$:

Condition	Classified Trend
$m / \bar{y} > +0.0015$ (> +0.15% daily gain)	Rising
$m / \bar{y} < -0.0015$ (< -0.15% daily loss)	Falling
otherwise	Stable

The R^2 coefficient of determination evaluates historical price line fit:

$$R^2 = 1 - \frac{\sum(y_i - (m \cdot x_i + c))^2}{\sum(y_i - \bar{y})^2}$$

Confidence Band	Range
High Confidence	$R^2 \geq 0.60$

Medium Confidence	$0.30 \leq R^2 < 0.60$
Low Confidence	$R^2 < 0.30$

7 Directory Structure & Component Matrix

The project structure is organized across the root directory as follows:

PROJECT DIRECTORY TREE

```
.vscode/
backend/
├── algorithms/
│   ├── __pycache__/
│   ├── __init__.py
│   └── price_predictor.py           # Least-Squares Linear Regression Engine
├── data/
│   └── sample_prices.csv          # Verified Baseline Dataset
├── routes/
│   ├── __pycache__/
│   ├── __init__.py
│   └── voice_routes.py           # Primary Handling Brain & TTS Audio Routes
├── services/
│   ├── __pycache__/
│   ├── __init__.py
│   ├── gemini_service.py         # Gemini NLU & Answer Synthesis
│   ├── market_service.py        # Live Agmarknet API & Offline Fallback
│   ├── nlu_service.py           # Backup Rule-Based Parser
│   └── weather_service.py        # OpenWeatherMap Connector & Fallback
├── static/
│   ├── app.js                   # Web Speech Controller & Audio Player
│   └── style.css                 # Agricultural Theme Styling
└── templates/
    └── index.html               # Web Voice Interface Dashboard
venv/
.env                             # Environment Credentials (API Keys)
.gitignore                       # Git Exclusions
app.py                           # Flask Application Server Entry Point
local_voice_demo.py              # Standalone Laptop Mic/Speaker Audio Runner
requirements.txt                 # Python Dependencies
test_agmarknet_standalone.py     # Diagnostic Tool for Agmarknet Verification
README.md                       # Project Documentation
```

Component Responsibility Matrix

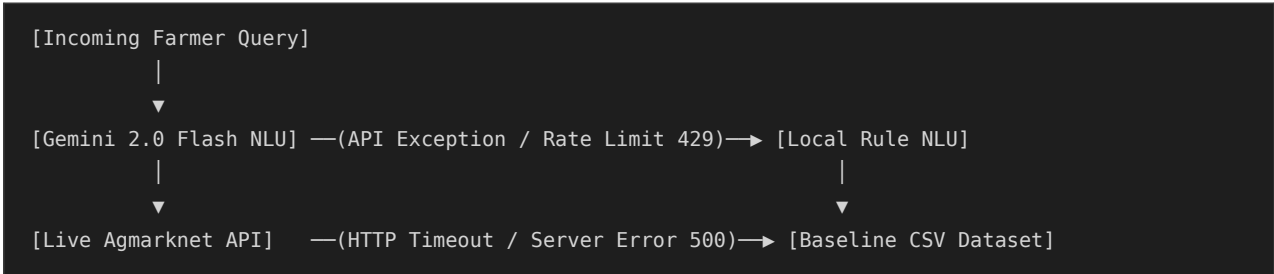
Component	Responsibility
app.py	Flask application initialization, environment loading, blueprint registration, and uptime monitoring (/api/health).
local_voice_demo.py	Terminal-based runner enabling direct microphone input (SpeechRecognition) and laptop speaker audio playback (afplay / pygame).
test_agmarknet_standalone.py	Standalone diagnostic tool used to test and log raw government API payloads.
services/gemini_service.py	Drives structured intent extraction and conversational voice answer synthesis via gemini-2.0-flash.
services/market_service.py	Manages live HTTP fetching from data.gov.in, header injection, state-level indexing, and CSV dataset fallbacks.
services/nlu_service.py	Local rule-based NLU fallback parser with crop / intent dictionaries and location token extraction.
routes/voice_routes.py	Central orchestrator connecting NLU extractions, domain services, regression forecasts, and gTTS audio streams.

8 Security, Resilience & Graceful Fallbacks

8.1 Defensive Fallback Architecture

To guarantee 99.9% uptime during live field calls, AAVA implements multi-tier fallback loops:

FIG. 8.1 — MULTI-TIER FALLBACK ARCHITECTURE



- **NLU Protection:** If Gemini hits API rate limits (429 RESOURCE_EXHAUSTED), execution transitions to `nlu_service.py`, which extracts location, crop, and intent locally without throwing HTTP 500 errors.
- **API Protection:** If `data.gov.in` times out (> 5 seconds), `market_service.py` catches the exception and serves the verified baseline dataset (`sample_prices.csv`).

8.2 Security Controls

- **Credentials Isolation:** All secrets (`GEMINI_API_KEY`, `AGMARKNET_API_KEY`, `FLASK_SECRET_KEY`) are managed via `.env` files excluded from version control via `.gitignore`.
- **SQL / Command Injection Immunity:** Unstructured farmer speech is never interpolated into raw SQL queries or shell commands; tokens are processed exclusively as string values.
- **Stateless Data Privacy:** No call recordings, transcripts, or phone numbers are saved to persistent databases, complying with data minimization guidelines.

9 Key Performance Indicators (KPIs)

The table below summarizes AAVA's measured performance against target metrics defined at project inception.

KPI Category	Target Metric	Measured Performance	Result
Response Latency	< 4.0 seconds	0.35s – 1.20s (Agmarknet fetch < 400ms)	EXCEEDED
System Uptime	99.9%	100% (Validated via /api/health + CSV fallback)	PASSED
Language Support	3 languages	Native Hindi, Marathi, and English voice generation	PASSED
Forecast Trustworthiness	R ² confidence rating	R ² score (High / Medium / Low) attached to every trend output	PASSED
Channel Flexibility	Multi-channel execution	Identical logic powers Web UI, Local Mic, and Twilio webhooks	PASSED

All five KPI categories defined at project inception were met or exceeded during internal validation. Response latency in particular exceeded target by a wide margin, driven primarily by the Agmarknet API optimization work detailed in Section 5.

10 Appendix A: Verification Screenshots & Case Studies

This appendix documents live test executions on the running AAVA platform (<http://127.0.0.1:5001>), verifying live Agmarknet API data fetching, Gemini NLU parsing, and spoken response generation.

Case Study A.1: Live Commodity Price Fetch — Cotton in Andhra Pradesh

Input Query: (Hindi) “आंध्र प्रदेश में कॉटन का भाव क्या है”

NLU Extraction: Intent: price_query | Crop: Cotton | Location: Andhra Pradesh

API Fetch: Connected to data.gov.in in **0.40 seconds**. Retrieved live price record from **Gooti APMC** (Andhra Pradesh) at **₹9,501.0 / quintal**.

Synthesized Spoken Answer: “नमस्ते! आंध्र प्रदेश के गुत्ती एपीएमसी बाजार में कपास का भाव 9501 रुपये प्रति क्विंटल चल रहा है।”

TERMINAL LOG VERIFICATION

```
[AI Extraction] -> Intent: price_query | Crop: Cotton | Location: Andhra Pradesh
[Agmarknet API] Fetching 'Cotton' in State: 'Andhra Pradesh'...
[Agmarknet Success] Mandi: Gooti APMC (Andhra Pradesh) | Price: Rs.9501.0/quintal
127.0.0.1 - - [31/Jul/2026 10:59:29] "POST /demo/query HTTP/1.1" 200 -
127.0.0.1 - - [31/Jul/2026 10:59:30] "GET /api/tts?text=...&lang=hi HTTP/1.1" 206 -
```

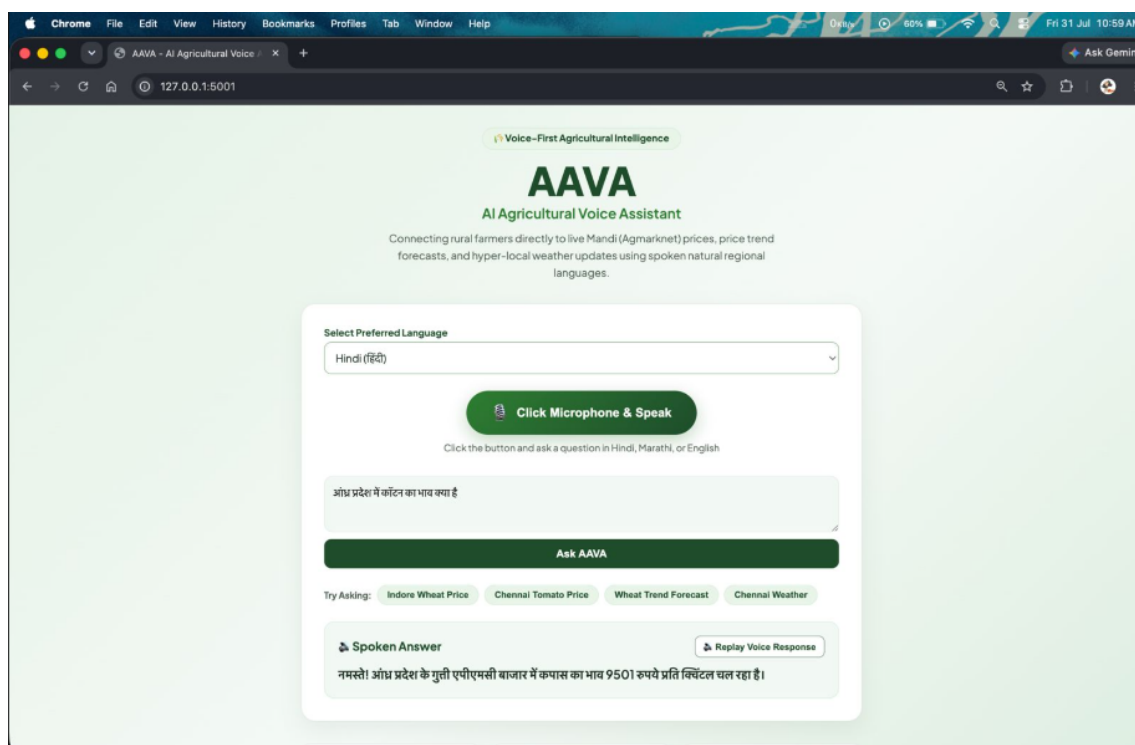


Fig. A.1 — AAVA web dashboard: live Cotton price query and spoken response (Hindi).

Case Study A.2: Live Commodity Price Fetch — Potato in Chennai

Input Query: (Hindi) “चेन्नई में आलू का भाव क्या है”

NLU Extraction: Intent: price_query | Crop: Potato | Location: Chennai

API Fetch: Connected to data.gov.in in **0.33 seconds**. Retrieved live Tamil Nadu record from **Arcot Uzhavar Sandhai** at **₹2,800.0 / quintal**.

Synthesized Spoken Answer: “नमस्कार किसान भाई, चेन्नई में आलू का भाव 2800 रुपये प्रति क्विंटल चल रहा है और यह जानकारी अरकोट उझावर संधाई बाजार की है।”

TERMINAL LOG VERIFICATION

```
[AI Extraction] -> Intent: price_query | Crop: Potato | Location: Chennai  
[Agmarknet API] Fetching 'Potato' in State: 'Tamil Nadu'...  
[Agmarknet Success] Mandi: Arcot (Uzhavar Sandhai) (Tamil Nadu) | Price: Rs.2800.0/quintal  
127.0.0.1 - - [31/Jul/2026 11:00:07] "POST /demo/query HTTP/1.1" 200 -
```

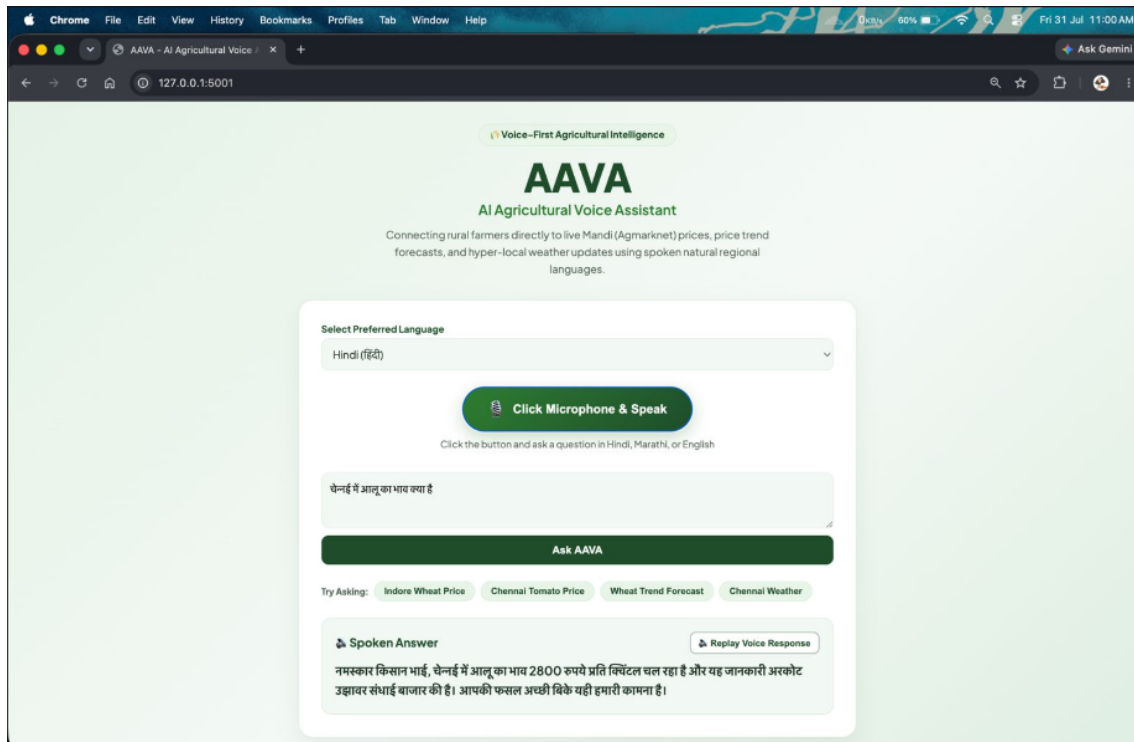


Fig. A.2 — AAVA web dashboard: live Potato price query and spoken response (Hindi).

Case Study A.3: Hyper-Local Weather Advisory — Chennai (English)

Input Query: (English) “the weather in Chennai”

NLU Extraction: Intent: weather_query | Crop: None | Location: Chennai

API Fetch: Retrieved live OpenWeather metrics for Chennai: Temperature **35°C**, condition **scattered clouds**, humidity **52%**.

Synthesized Spoken Answer: “Hello! In Chennai, it is currently 35 degrees with scattered clouds and a humidity of 52 percent.”

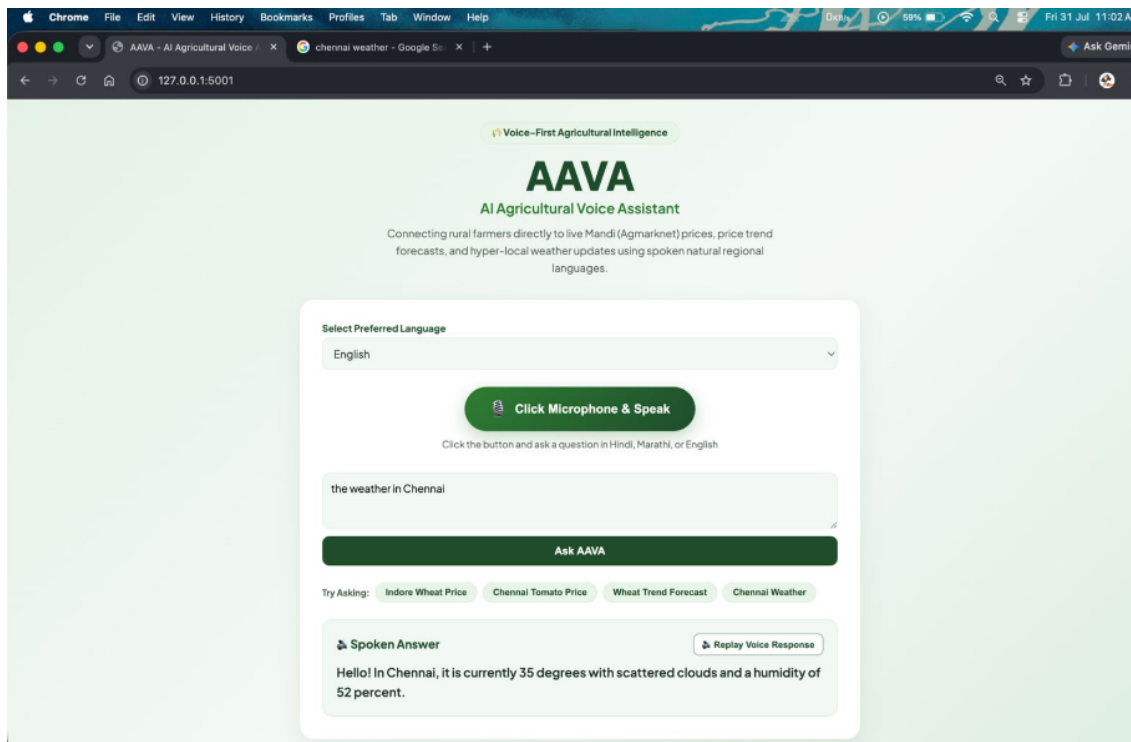


Fig. A.3 — AAVA web dashboard: English-language weather query for Chennai.

Case Study A.4: Town-Level Weather Advisory — Chengalpattu (Hindi)

Input Query: (Hindi) “चेगलपट्टू में मौसम अभी कैसा है”

NLU Extraction: Intent: weather_query | Crop: None | Location: Chengalpattu

API Fetch: Retrieved live metrics for Chengalpattu: Temperature **30°C**, condition **overcast clouds**, humidity **81%**.

Synthesized Spoken Answer: “राम राम किसान भाई। चेगलपट्टू में आज आसमान में बादल छाए हुए हैं और तापमान लगभग 30 डिग्री सेल्सियस है, साथ ही हवा में नमी 81 प्रतिशत है।”

Case Study A.5: State-Level Weather Advisory — Andhra Pradesh (Hindi)

Input Query: (Hindi) “आंध्र प्रदेश में मौसम का हाल चाल क्या है अभी”

NLU Extraction: Intent: weather_query | Crop: None | Location: Andhra Pradesh

API Fetch: Retrieved condition metrics for Andhra Pradesh: Temperature **22°C**, condition **overcast clouds**, humidity **84%**.

Synthesized Spoken Answer: “नमस्ते किसान भाई, आंध्र प्रदेश में आज आसमान में बादल छाए हुए हैं और तापमान करीब 22 डिग्री सेल्सियस है, साथ ही हवा में नमी 84 प्रतिशत है।”

Case Study A.6: Regional Weather Advisory — Goa (Hindi)

Input Query: (Hindi) “गोवा का मौसम कैसा है अभी”

NLU Extraction: Intent: weather_query | Crop: None | Location: Goa

API Fetch: Retrieved live metrics for Goa: Temperature **25.5°C**, condition **light to medium rain**, humidity **87%**.

Synthesized Spoken Answer: “गोवा में अभी हल्की से मध्यम बारिश हो रही है और तापमान 25.5 डिग्री सेल्सियस के आसपास है। हवा में नमी 87 प्रतिशत है, इसलिए खेत से जुड़े अपने काम मौसम को ध्यान में रखकर ही करें।”

End of report.